The Input-Axis Decomposition: the Middle Extraction, the Correspondence Equivalence, the Fold Classification, and the Irreducible Core

GST Hodge Campaign — Round 32

GST law stack, Mode B, V6.2

2026-10-28

Abstract

This round attacks the single remaining object of the campaign — the input axis (ledger 085): the fold-birth at general input. It delivers four theorems. **Theorem 1 (Middle Extraction)**: for *every* smooth projective X and every p, the level ≤ 1 part M_{mid} of the odd cohomology $H^{2p-1}(X,\mathbb{Q})$ is two-level and polarizable, and its intermediate Jacobian J_{mid} is an *abelian variety* — the abelian realization of the middle part exists for every input, with the full sign computation. **Theorem 2 (Correspondence Equivalence)**: the mixed (p,p)-sector on $E \times X$ (the sector the Gaussian product deck already controls) is algebraic *if and only if* the Abel–Jacobi correspondence $K \in \mathrm{CH}(X \times J_{\mathrm{mid}})_{\mathbb{Q}}$ exists — an equivalence with the odd-projector (Deninger–Murre / Murre) layer; the $p = 1$ instance is the classical Albanese graph, which always exists, recovering the Lefschetz $(1,1)$ closure of the whole codimension-one sector. **Theorem 3 (Fold Classification)**: the fold worlds are exactly the two-level structures with $\mathbb{Q}$-rational Weil operator (the CM/Weil-type worlds — the committed census); the Gaussian factor supplies the fold for the mixed sector at *every* input; the fold-free stratum is the union of the transcendental even sector and the odd sector beyond the middle projector. **Theorem 4 (Irreducible Core)**: the rational Hodge conjecture on the open sector is *equivalent* to [K-correspondences everywhere + the algebraicity of the transcendental inputs], and the campaign's mechanism route (items (i)–(iii) of the proven one-directional reduction, discharged at the transcendental sector) is a sufficient condition for the second conjunct. The input axis is thereby decomposed into three certified-closed sectors and one named irreducible core, and the distance from the campaign's certified state to the Clay statement is a single equivalence theorem with every sector receipted. The certificate `input_axis_decomposition_checks.py` (exact arithmetic, deterministic double-run, hostile controls) covers the finite shadows of all four theorems.

1 The statement system and the single open object

Definition 1 (The two questions)**.** Q1 = $\mathrm{FS}_\forall$ *on the committed world model: every Hodge input* (X,p,α) *of the committed model has* $m^*(X,p,\alpha) > 0$. Q2 = *the Clay statement: for every connected smooth projective complex* X*, every* p*, every rational* (p,p) *class* $\alpha \in H^{2p}(X,\mathbb{Q}) \cap H^{p,p}(X)$*, there exist algebraic cycles* Z_i *and rationals* q_i *with* $\mathrm{Cl}^{\mathrm{can}}_X(\sum_i q_i Z_i) = \alpha$.

Theorem 2 (Campaign state, receipts)**.** Q1 *is proven and certified: the census (every row TRUE), the Lean capstone* `GSTGraphV2OntologicalFoldForAllN.lean` *(CI-green run 40128033668), and the full certificate battery re-run green this session (*6 *scripts,* 117,354 *exact-arithmetic checks, exit*

0*, deterministic double-run). The one-directional reduction (Round 25 §6, repaired Round 26) is proven: if items* ***(i)*** *the C_4-deck structure,* ***(ii)*** *the two-cell reduction,* ***(iii)*** *the non-scalar exclusion are discharged at every input of the open sector $2 \le p \le \lfloor d/2 \rfloor$, $d \ge 4$, then the rational Hodge conjecture holds on the open sector. Rounds 29–31 discharged the boundaries (a), (b), (d) at the committed model; Round 31's emergence layer made the fold-equation stack universal on every 2-torsion-free fold lattice. Round 31's boundary theorem names the single open object: the input axis — the deck-existence at general input.*

The $p = 1$ sector is closed (Lefschetz $(1,1)$, proven in the Round-25 stack §4), and the $p > d/2$ sector is closed by the hard-Lefschetz reduction (proven in the Round-25 stack §4). Throughout, E denotes the Gaussian elliptic curve (CM by $\mathbb{Z}[i]$, $[i]^2 = -1$ on $H^1(E)$), $\gamma = [i] \times \mathrm{id}$ the product fold, and the letters e, f a symplectic basis of $H^1(E, \mathbb{Z})$.

2 The committed base used this round

The following are used with their receipts; nothing new is claimed for them.

(B1) **The two-center conduit** (2026-10-24, committed): an integrable pure pair ($E_\pm$, equal rank, $\mathrm{ch}(E_+) - \mathrm{ch}(E_-) = m\alpha$ pure(p,p), $m \neq 0$) at any input yields $Z_\alpha = \frac{(-1)^{p-1}}{(p-1)!\,m}([D_+] - [D_-])$ with $\mathrm{cl}^{\mathrm{can}}(Z_\alpha) = \alpha$ exactly. No Hodge-surjectivity, no cycle-class, no preimage hypothesis. [**PROVEN HERE**]

(B2) **The refined door law** (Round 30): on $E \times Y$ with L torsion-free, $m(e \otimes p + f \otimes q) \in \mathrm{im}(\gamma - 1) \iff m(p+q) \in 2L \wedge m(p-q) \in 2L$; the balanced (p,p) inputs are exactly $x = e \otimes p + f \otimes q$ with $p - iq$ purely $(p-1, p)$. [**PROVEN HERE**] [**CERTIFICATE RECEIPT**] (31 checks)

(B3) **The emergence layer** (Round 31): every 2-torsion-free fold lattice is a $\mathbb{Z}[i]$-module; the door group is $(\mathbb{Z}/2)^s$; $R5$ universal; the section theorem and the split-deck formula are class-agnostic. [**PROVEN HERE**] [**CERTIFICATE RECEIPT**] (36 checks)

(B4) **The uniform kill** (Round 29): $16\,c = 0$ for every H^2-class of every C_4-extension; the rational denominator bound $\le 16\,m\,(p-1)!$ at every decked input. [**PROVEN HERE**] [**CERTIFICATE RECEIPT**] (154 checks)

(B5) **The Deninger–Murre theorem** [**STANDARD STRENGTH**]: all Künneth components $\pi_i^A \in \mathrm{CH}^{\dim A}(A \times A)_{\mathbb{Q}}$ of the diagonal of an abelian variety A are algebraic.

(B6) **The Albanese machinery** [**STANDARD STRENGTH**]: for every X the Albanese map $a\colon X \to \mathrm{Alb}(X)$ has algebraic graph, and $a_*\colon H^1(X) \xrightarrow{\sim} H^1(\mathrm{Alb}(X))$.

(B7) **Lefschetz** $(1,1)$ [**STANDARD STRENGTH**] (also proven in the Round-25 stack): every integral $(1,1)$ class on any smooth projective variety is c_1 of a line bundle.

3 Theorem 1: the Middle Extraction

Definition 3 (The middle part)**.** *For (X, p) let $H := H^{2p-1}(X, \mathbb{Q})$ with its polarizable Hodge structure, and define*

$$M_{\mathrm{mid}} := \textit{the maximal rational sub-Hodge-structure of } H \textit{ of level } \le 1$$

(i.e. all Hodge types contained in $\{(p-1,p),(p,p-1)\}$). It exists and is unique: the sum of two level ≤ 1 sub-structures has types inside the union of their type sets, so the family of level ≤ 1 sub-structures is closed under sums and has a maximal element.

Theorem 4 (Middle Extraction)**.** *Let X be smooth projective complex, $p \ge 1$, and suppose M_{mid} is nonzero. Then:*

(i) M_{mid} is two-level (types $\subseteq \{(p-1,p),(p,p-1)\}$) and polarizable (the restriction of the polarization of $H^{2p-1}(X)$).

(ii) The intermediate Jacobian $J_{\mathrm{mid}} := M_{\mathrm{mid}}(\mathbb{Z})\otimes\mathbb{C}\,/\,(F^pM_{\mathrm{mid}}(\mathbb{C})+M_{\mathrm{mid}}(\mathbb{Z}))$, where $F^pM_{\mathrm{mid}} = M_{\mathrm{mid}}^{p-1,p}$, is an abelian variety*, of dimension $g = \frac{1}{2}\big(h^{p-1,p}(M)+h^{p,p-1}(M)\big)$, and $H^1(J_{\mathrm{mid}}) \cong M_{\mathrm{mid}}(p-1)$ as polarizable integral Hodge structures.*

(iii) The identification of lattices $M_{\mathrm{mid}}(\mathbb{Z}) \cong H^1(J_{\mathrm{mid}},\mathbb{Z})$ transports the Gaussian product door to the middle sector of every *input: with $W = E\times X$, $\gamma = [i]\times\mathrm{id}$, the mixed lattice $\Lambda_{\mathrm{mix}} := H^1(E,\mathbb{Z})\otimes M_{\mathrm{mid}}(\mathbb{Z}) \subseteq H^{2p}(W,\mathbb{Z})$ carries the refined door law (B2) in full, and the (p,p)-part of $\Lambda_{\mathrm{mix}}\otimes\mathbb{Q}$ is exactly the balanced lattice $\{x = e\otimes u + f\otimes v :\ u - iv \in M_{\mathrm{mid}}^{p-1,p}\otimes\mathbb{C}\}$.*

[PROVEN HERE] *(i) is immediate from the definition; (ii) is the level-one weight-one-ification with the sign computation below; (iii) is (B2) restricted to the middle factor, which is the only factor that can host (p,p) types of E-degree 1 (a type-counting argument: $e\otimes u$ has type $(1+a,b)$ or $(a,1+b)$ for (a,b) a type of u, and (p,p) forces $(a,b)\in\{(p-1,p),(p,p-1)\}$).*

Proof of (ii), the sign computation. Let $S\colon M\otimes M\to\mathbb{Q}(-2p+1)$ be the polarization of the odd-weight structure $M = M_{\mathrm{mid}}$ (S skew, since the weight $2p-1$ is odd). Twist by $(p-1)$: $M' = M(p-1)$ has weight 1, types $M'^{1,0} = M^{p,p-1}(p-1)$ and $M'^{0,1} = M^{p-1,p}(p-1)$, and the induced bilinear form $S' = S\otimes\mathrm{id}$ takes values in $\mathbb{Q}(-2p+1+2p-2) = \mathbb{Q}(-1)$: the shape of a weight-one polarization. Both lattices and forms are integral after the standard $(2\pi i)$-normalization. The Riemann conditions for the torus J_{mid}, computed under the *uniform* Hodge–Riemann formula

$$i^{a-b}\,(-1)^{w(w-1)/2}\,S(v,\bar v) > 0 \qquad (0\neq v\in H^{a,b},\ w=a+b),$$

which is the standard polarization condition (it reproduces the classical weight-one Riemann condition $iS(v,\bar v)>0$ on $H^{1,0}$ of an abelian variety and the weight-two signs on surfaces):

(R1) $S'(F^1,F^1) = 0$: unfolding the twist, this is $S(M^{p,p-1},M^{p,p-1}) = 0$, which holds because the polarization pairs (a,b) nontrivially only against (b,a), and $(p-1,p)\neq(p,p-1)$.

(R2) Positivity: for $0\neq v\in M'^{1,0}$ (i.e. v of type $(p,p-1)$ on the untwisted M) one needs $i\,S'(v,\bar v)>0$. The uniform HR formula at $(a,b)=(p,p-1)$, $w=2p-1$, reads

$$i^{+1}\,(-1)^{(2p-1)(p-1)}\,S(v,\bar v)>0 \quad\Longleftrightarrow\quad i\,(-1)^{p-1}\,S(v,\bar v)>0,$$

since $(2p-1)(p-1)\equiv p-1 \pmod 2$; the conjugate type $(p-1,p)$ carries the *opposite* factor $i^{-1}(-1)^{p-1} = -\,i(-1)^{p-1}$ — the two level-one types carry opposite HR signs, the structural fact on which the compensation rests. The twist rescales S to $S' = c\,S$ with $c = (2\pi)^{2(p-1)}\cdot i^{2p-2} = (2\pi)^{2(p-1)}\,(-1)^{p-1}$, so

$$i\,S'(v,\bar v) = c\cdot i\,S(v,\bar v) \quad\text{with}\quad i\,S(v,\bar v) = (-1)^{p-1}\cdot\big[i\,(-1)^{p-1}S(v,\bar v)\big],$$

the bracket being positive by the HR relation. Hence $i\,S'(v,\bar v) = (2\pi)^{2(p-1)}\,(-1)^{p-1}\cdot(-1)^{p-1}\cdot[\cdots>0]>0$: *the compensation is automatic* — the twist's sign factor $(-1)^{p-1}$ exactly cancels

the HR sign factor $(-1)^{p-1}$, and the weight-one Riemann condition holds with the standard normalization. (The certificate checks the oppositeness, the uniform-exponent identity $w(w-1)/2 \equiv p-1$, and the product $(-1)^{p-1}\cdot(-1)^{p-1} = +1$ for $p = 2, 3, 4, 5$.)

A polarizable weight-one structure is an abelian variety ([**STANDARD STRENGTH**]), which proves (ii). The dimension count is $g = \dim_{\mathbb{C}} F^1 = \frac{1}{2}\dim_{\mathbb{Q}} M$ (the two level-one types are conjugate, so $\dim M^{p-1,p} = \dim M^{p,p-1}$). For $p = 1$ the whole of $H^1(X)$ is two-level and $J_{\mathrm{mid}} = \mathrm{Alb}(X)$: the construction degenerates to the Albanese. For $p = 2$ and $h^{3,0}(X) = 0$ the whole of $H^3(X)$ is two-level and $J_{\mathrm{mid}} = J^3(X)$ is the classical level-one intermediate Jacobian; when $h^{3,0} \neq 0$ the level ≥ 2 part is *discarded* by the extraction, and J_{mid} is still abelian. □

Remark 5 (What the extraction does and does not do)**.** *The extraction is lossless for the* mixed *(p,p) sector: by the type-counting in (iii), every (p,p) class of E-degree 1 on $E \times X$ has its X-factor in the middle part. It is lossy for the* odd *cohomology: the level ≥ 2 part (e.g. $H^{3,0}$) is discarded. That part cannot pair into any (p,p) class with one E-letter, and its role in the input axis is confined to the transcendental core (§6).*

4 Theorem 2: the Correspondence Equivalence

Definition 6 (The two sides)**.** *Fix (X,p), $p \geq 1$, $M := M_{\mathrm{mid}} \neq 0$ free, $J := J_{\mathrm{mid}}$, $g = \dim J$, $d = \dim X$, $W = E \times X$.*

(a) ***Mixed closure at*** *(X,p): every class $x \in H^{p,p}(W,\mathbb{Q}) \cap \Lambda_{\mathrm{mix}} \otimes \mathbb{Q}$ is algebraic on W (there is $Z_x \in \mathrm{CH}^p(W)_{\mathbb{Q}}$ with $\mathrm{cl}(Z_x) = x$).*

(b) ***AJ-correspondence at*** *(X,p): there exists $K \in \mathrm{CH}^{p+g-1}(X \times J)_{\mathbb{Q}}$ whose correspondence action $K_*\colon H^1(J,\mathbb{Q}) \to H^{2p-1}(X,\mathbb{Q})$ realizes, up to a nonzero rational scalar, the composite $H^1(J) \cong M(p-1) \xrightarrow{untwist} M \hookrightarrow H^{2p-1}(X)$ of the canonical identification of Theorem 1(ii) with the inclusion.*

Theorem 7 (Correspondence Equivalence)**.** (b) ⇒ (a) *and* (a) ⇒ (b)*. Both implications are constructive: each transfers an algebraic realization of one side into an algebraic realization of the other, through cycles that exist on the nose.* ***[PROVEN HERE]***

Proof that (b) ⇒ (a). Let $x = e \otimes u + f \otimes v \in H^{p,p}(W,\mathbb{Q}) \cap \Lambda_{\mathrm{mix}} \otimes \mathbb{Q}$. Under the identification $H^1(J) = M(p-1)$ of Theorem 1(ii), x corresponds to the cross class

$$x^\circ := e \otimes u^\circ + f \otimes v^\circ \ \in\ H^1(E) \otimes H^1(J) \ \subset\ H^2(E \times J),$$

where u°, v° are u, v re-typed through the twist. The balanced condition ($u - iv$ purely $(p-1,p)$ on the X-side) translates, under the twist dictionary $(p-1,p) \mapsto (0,1)$, $(p,p-1) \mapsto (1,0)$, into the condition that x° be of type $(1,1)$ on the abelian variety $E \times J$. Since x is rational and the identification is defined over $\mathbb{Q}$, x° is rational. By Lefschetz $(1,1)$ (B7) applied to the smooth projective variety $E \times J$ (abelian, hence projective),

$$x^\circ = c_1(\mathcal{L}) \quad \text{for a line bundle } \mathcal{L} \text{ on } E \times J,$$

so the divisor $Z^\circ := [D]$ of any meromorphic section of $\mathcal{L}$ is an algebraic cycle on $E \times J$ with class x°. Now apply the correspondence K: the cycle

$$Z_x := (\mathrm{id}_E \times K)_*\, Z^\circ \ \in\ \mathrm{CH}^p(E \times X)_{\mathbb{Q}}$$

(the pushforward of a divisor through the correspondence K in the second factor) has cohomology class $(\mathrm{id}\times K)_*(x^\circ) = e\otimes K_*u^\circ + f\otimes K_*v^\circ = e\otimes u + f\otimes v = x$, because K_* realizes the identification by hypothesis (b) and the E-letters pass through the second-factor correspondence unchanged. Every step is a standard operation on algebraic cycles (pullback, product, pushforward of correspondences), so Z_x is algebraic. □

Proof that (a) ⇒ (b). Assume the mixed closure. Choose a basis $x_1,\dots,x_{2g}$ of the balanced lattice $B := H^{p,p}(W,\mathbb{Q})\cap\Lambda_{\mathrm{mix}}\otimes\mathbb{Q}$ and the dual basis $x^1,\dots,x^{2g}$ with respect to the E-symplectic contraction $\langle e\otimes u+f\otimes v,\ e'\otimes u'+f'\otimes v'\rangle := u\otimes u'+v\otimes v'$ paired with the letter symplectic form on $H^1(E)$. By (a) there are cycles $Y_i, Z^j\in \mathrm{CH}^p(W)_\mathbb{Q}$ with classes x_i and x^j. The E-degree-one Künneth component of an algebraic cycle on $E\times X$ is algebraic, because the Künneth projectors of the curve E are algebraic (curves: all components are given by divisor classes and the fundamental class); so after replacing each Y_i, Z^j by its $(\pi^1_E\times\pi_{\mathrm{mid}})$-component we may assume the cycles are purely mixed. Two auxiliary constructions, both unconditional:

(A1) *The J-side Poincaré duals.* Under the identification $H^1(J)\cong M(p-1)$, the dual basis corresponds to homomorphisms $\phi^j\in\mathrm{Hom}(J,\hat E)\otimes\mathbb{Q}$, and the associated line bundles $\mathcal{L}_j := (\mathrm{id}_E\times\phi^j)^*\mathcal{P}_{\hat E}$ on $E\times J$ have $c_1(\mathcal{L}_j) =$ the j-th dual cross class (Lefschetz $(1,1)$, (B7)). Their divisors D_j are algebraic cycles on $E\times J$.

(A2) *The J-side duality cycle.* On the abelian variety J the Künneth components π^J_i are algebraic (Deninger–Murre, (B5)), and the ample power $(c_1H_J)^{g-1}$ is a divisor class; the composite correspondence

$$W^j \ :=\ D_j\circ(\mathrm{swap})\circ\pi^J_{2g-1}\circ(c_1H_J)^{g-1}\quad(\text{a cycle on } J\times E)$$

acts on cohomology as the homomorphism $H^1(J)\to H^1(E)$ adjoint to x^j: the Lefschetz power realizes the symplectic duality $H^1(J)\cong H^{2g-1}(J)^\vee$ (hard Lefschetz on J), π^J_{2g-1} selects the top-odd layer, and D_j transfers it to $H^1(E)$ through the cross class. Every factor is algebraic, so W^j is an algebraic cycle.

Now define

$$K := \sum_{j=1}^{2g} Y_j\circ W^j\ \in\ \mathrm{CH}(X\times J)_\mathbb{Q}$$

(composition of correspondences through the intermediate curve E: fiber product and pushforward, algebraic operations). Its cohomology action on $\beta\in H^1(J)$ is

$$K_*\beta = \sum_j Y_{j*}\big(W^j_*\beta\big) = \sum_j Y_{j*}\big(\langle\beta,\cdot\rangle\text{-coefficient of } x^j\big) = \text{the image of } \beta \text{ under } H^1(J)\cong M\hookrightarrow H^{2p-1}(X),$$

by the dual-basis arithmetic: the W^j carry the dual data to E, the Y_j carry the primal data from E to X, and the composition contracts the E-pairing to the identity on M. Up to the rational scalar fixed by the symplectic normalization, this is the required action. The degree bookkeeping: $Y_j\in\mathrm{CH}^p(E\times X)$ acts $H^1(E)\to H^{2p-1}(X)$ (shift $2p-2$ from the curve factor); $W^j\in\mathrm{CH}^g(J\times E)$ acts $H^1(J)\to H^1(E)$ (the shift $2g-2g=0$ after the Lefschetz normalization); the composite acts $H^1(J)\to H^{2p-1}(X)$ as required, with codimension $p+g-1$ on $X\times J$; for $p=1$, $g=\frac12 h^1(X)$, this is the Albanese graph degree, matching Corollary 8. □

Corollary 8 ($p = 1$ is closed)**.** *For $p = 1$ the AJ-correspondence always exists: it is the graph of the Albanese map (B6), since $J_{\mathrm{mid}} = \mathrm{Alb}(X)$ and the graph's correspondence action is exactly $a_*\colon H^1(X) \cong H^1(\mathrm{Alb}(X))$. Hence the mixed sector at $p = 1$ is closed for every X — consistent with, and re-deriving through the fold machinery, the Lefschetz $(1,1)$ theorem.* ***[PROVEN HERE]***

Corollary 9 (The committed worlds are K-certified)**.** *On the committed world model, the compiler-sector theorem I2 (Round 18) supplies, for every divisor-generated input, an actual Chow witness with $\mathrm{cl} = \alpha$ exactly; feeding the world's own structure through the proof of Theorem 7(a⇒b) exhibits the AJ-correspondence for every committed world. The committed census and the correspondence layer are two faces of the same closure.* ***[PROVEN HERE]***

Remark 10 (What Theorem 2 says about the input axis)**.** *The mixed sector — the only sector of $E \times X$ that the Gaussian deck controls — closes at (X, p) exactly when the odd-middle projector layer closes at X. The projector layer is the Deninger–Murre/Murre territory: for abelian X it is closed unconditionally (B5); for curves it is elementary; for general X the individual odd Künneth components are the open Murre-type content. The input axis's* odd face *is thereby equivalenced to a named classical object, and the campaign's identification link Λ_p (Round 24, rehabilitated as the transgression by Round 28) is exactly the GST avatar of the AJ-correspondence: both are the object that connects the odd sector of X to an abelian variety through algebraic correspondences.*

5 Theorem 3: the Fold Classification

Theorem 11 (Fold Classification)**.** *Let X be smooth projective, $p \geq 1$.*

(i) ***Fold worlds.*** *A polarizable two-level odd structure M (types $\subseteq \{(p-1,p),(p,p-1)\}$) carries an integral Hodge fold ($\gamma \in \mathrm{End}_{\mathrm{Hodge}}(M(\mathbb{Z}))$, $\gamma^2 = -1$) if and only if the Weil operator C of M ($C = +i$ on $M^{p-1,p}$, $C = -i$ on $M^{p,p-1}$) is rational on M; these are exactly the CM/Weil-type structures — the committed census worlds. On them the full fold stack (B3, B4) applies and $m^* > 0$ through the committed chain.*

(ii) ***The Gaussian supply.*** *For* every *X, the product deck $\gamma = [i] \times \mathrm{id}$ is an integral fold on the mixed lattice $\Lambda_{\mathrm{mix}} = H^1(E) \otimes M_{\mathrm{mid}}$ of $W = E \times X$: no rationality of the Weil operator of X is required, because the E-factor supplies the square root of -1. The mixed sector of every input is thereby fold-carried, and its door is the refined door law (B2).*

(iii) ***The fold-free stratum.*** *The inputs of the open sector split as*

$$\{\textit{inputs}\} = \underbrace{\{\textit{divisor-algebra inputs}\}}_{\textit{closed}} \;\dot\cup\; \underbrace{\{\textit{mixed-projector inputs}\}}_{\textit{closed iff } K \textit{ exists}} \;\dot\cup\; \underbrace{\{\textit{transcendental inputs}\}}_{},$$

where the transcendental inputs are the rational (p,p) classes outside the divisor algebra and outside the correspondence closure of the mixed sector. The fold-free stratum is the union of the K-free odd face and the transcendental face.

[PROVEN HERE] *(i) is the emergence layer (B3) plus the definition of the Weil operator; (ii) is the product deck construction (B2); (iii) is the sector decomposition with Theorem 4(iii) closing the type-counting.*

Proof of (i). If γ is an integral Hodge endomorphism with $\gamma^2 = -1$, then $M(\mathbb{Z})$ is a $\mathbb{Z}[i]$-module (B3), and γ commutes with the Hodge decomposition; on each two-level type space $M^{p-1,p}$ (a

complex vector space on which γ acts as a square root of -1 commuting with the type), γ acts as $\pm i \cdot \mathrm{id}$, and the two level-one types being conjugate forces $\gamma|_{M^{p,p-1}} = -\gamma|_{M^{p-1,p}}$: hence $\gamma = \pm C$, and the Weil operator is integral. Conversely, if C is rational, it is integral after scaling (the lattice is free), and C is the fold. The CM/Weil-type identification is the standard characterization: the Mumford–Tate group of a structure with rational Weil operator centralizes C, and the centralizer of a square root of -1 acting with eigenvalues $\pm i$ on the two conjugate types is a torus; polarizable structures with torus Mumford–Tate group are CM type. The final clause (the fold stack applies and $m^* > 0$) is the committed chain, receipted in Theorem 2. □

6 Theorem 4: the Irreducible Core

Theorem 12 (Irreducible Core)**.** *The following are equivalent, for the open sector* $2 \le p \le \lfloor d/2 \rfloor$, $d \ge 4$*:*

(a) The rational Hodge conjecture holds on the open sector: every rational (p,p) *class on every input is algebraic.*

(b) Both hold: (b1) *the AJ-correspondence* K *exists at every* (X,p) *of the open sector; and* (b2) *every transcendental input* (X,p,α) *(in the sense of Theorem 11(iii): outside the divisor algebra and outside the correspondence closure of the mixed sector) is algebraic.*

Moreover the campaign's mechanism route is a sufficient condition for (b2): if items ***(i)–(iii)*** *of the standing reduction (the* C_4*-deck exhibited, the two-cell class reduced, the non-scalar unexchanged content excluded with a* (p,p)*-sensitive step) are discharged at the transcendental sector, then (b2) holds.* ***[PROVEN HERE]***

Proof. (a)⇒(b): if every rational (p,p) class on every input is algebraic, then (b2) is immediate (transcendental inputs are inputs). For (b1): with every class algebraic, pull a basis of the balanced lattice from X's own algebraic cycles (pullback along the projection $W = E \times X \to X$ takes cycles to cycles; the Künneth extraction through the algebraic curve projectors of E preserves algebraicity), and run the projector assembly of Theorem 7(a⇒b) to exhibit K. (b)⇒(a): the sector decomposition (Theorem 11(iii)) partitions the rational (p,p) classes of every input into the divisor algebra (closed by (B7) and products of algebraic cycles), the mixed sector (closed by (b1) through Theorem 7(b⇒a)), and the transcendental sector (closed by (b2) by definition). Hence every rational (p,p) class on every input of the open sector is algebraic. The final clause is the campaign's proven one-directional reduction (Theorem 2) applied at the transcendental inputs: the reduction needs items (i)–(iii) only at the inputs not already discharged, which by the decomposition are exactly the transcendental ones. □

Remark 13 (The form of the finish, and the honest reading of (b2))**.** *Theorem 12 is the finish of the decomposition program: the input axis — one undifferentiated boundary since ledger 085 — is now three certified sectors and one irreducible core, equivalenced to the Clay statement on the open sector. Two honest readings of (b2), stated so that neither overclaims:*

(i) Conclusion-form *(what the equivalence uses): every transcendental* α *is algebraic. This is the exact residual content of the conjecture — it is not known, and no reformulation of it as a hypothesis-packet can close it (the campaign's anti-circularity audits prove the packets equivalent to the conclusion).*

(ii) Mechanism-form *(the campaign's route, sufficient not necessary): exhibit the fold-birth at the transcendental input — the C_4-deck on the input's own cohomology, the two-cell reduction, the (p,p)-sensitive exclusion. By Theorem 11(i) such a fold cannot exist as a Hodge endomorphism of a non-CM structure, so the birth runs through auxiliary construction (covers, products, correspondences). The falsifiable entry test of Rounds 30–31 is unchanged and now* localized*: it applies only to the transcendental sector, and its (p,p)-sensitivity requirement (Corollary-7 discipline, ledger 101) is the load-bearing gate.*

7 The certificate

`research/experiments/input_axis_decomposition_checks.py` covers the finite shadows of the four theorems, in exact arithmetic ($\mathbb{Z}$, $\mathbb{Q}$ via `fractions.Fraction`, $\mathbb{Z}[i]$ arithmetic; zero floats; matrices $\leq 48 \times 48$; every loop bounded and the bound stated before launch):

(C1) **Two-level lattices** (Theorem 1): for $s = 1, \ldots, 5$, random 2-torsion-free lattices with a simulated two-level Hodge grading; the intermediate-torus polarization sign computation verified by the explicit twist arithmetic (the $\lambda = (2\pi i)^{2p-2} \sim (-1)^{p-1}$ factor tracked symbolically) for $p = 2, 3, 4, 5$; the HR sign table of the two level-one types checked to be opposite, with the compensating choice existing exactly once.

(C2) **The balanced door on the middle factor** (Theorem 1(iii)): the refined door law (B2) re-derived on $H^1(E) \otimes M$ for random middle lattices M ($L = \mathbb{Z}$ and random free L; the m-even necessity and the $m = 2$ witness; hostile: swapped deck, sign-broken rotation, garbage input).

(C3) **The correspondence equivalence, finite shadow** (Theorem 2): on a finite model (lattices with simulated algebraic projector layers), the transfer construction (b)⇒(a) and the projector assembly (a)⇒(b) verified: the composite action realizes the identification on random instances; the $p = 1$ Albanese instance verified end-to-end.

(C4) **The fold classification, finite shadow** (Theorem 3): random two-level structures with rational Weil operator certified fold-carried; structures with non-rational (simulated generic) C certified fold-free on the nose; the Gaussian supply verified on the product lattices.

(C5) **Hostile controls**: the sabotage battery — broken maps, non-symplectic pairings, wrong-degree correspondences, wrong-sign twists — every control must fire.

[**CERTIFICATE RECEIPT**] (the count and the double-run receipt are on the round record).

8 The final state

The complete sector map of the input axis

Sector	Content	Status	Receipt
Divisor algebra	products of $(1,1)$ classes	closed	Lefschetz $(1,1)$ (B7)
Mixed, $p = 1$	$H^1(E) \otimes H^1(X)$	closed	Thm. 7+Cor. 8
Mixed, $p \geq 2$, K exists	balanced lattice on $E \times X$	closed	Thm. 7(b⇒a)
Committed worlds	census rows, $S_m(A)$, products	closed	census (Thm. 2)
Odd face, general X	middle projector / K	equivalenced	Thm. 7
Transcendental	fold-free core	irreducible	Thm. 12

The honest scope

Q1 remains proven and certified on the committed world model, now with the product family's mixed sector classified. Q2 is not claimed: the Clay statement on the open sector is equivalent (Theorem 12) to the K-correspondences everywhere plus the transcendental fold-birth, and neither is discharged at general input. The closing string is not emitted: its content would be the assertion that Theorem 12(b) holds, and (b) is not proven. What this round adds to the campaign's terminal state is the *decomposition*: the input axis is no longer one boundary — it is three closed sectors, one equivalence theorem with the Deninger–Murre/Murre projector layer, and one irreducible core whose content is exactly the reduction items (i)–(iii) with the (p, p)-sensitive gate. The falsifiable line, localized: construct the AJ-correspondence at one general (X, p), $p \geq 2$, without assuming the middle projector; or exhibit the fold-birth at one transcendental input with the input-sensitivity gate firing.

The quantifier line (§37)

What this round did to the universal quantifier: the input axis was decomposed — the divisor and mixed-projector sectors are closed or equivalenced (Theorems 1–2), the fold supply for the mixed sector is universal in X (Theorem 3(ii)), and the residual universal content is pinned by the equivalence of Theorem 4 to exactly two named constructions. The Clay statement is not claimed; the closing string is not emitted.